

Grade 8
Unit 4
Lesson 8 Practice

Name Answer Key
Date _____
Period _____

For questions 1-6, simplify each expression applying the laws of exponents.

1. $\frac{n^{12}}{n^4}$ n^8

2. $\frac{28v^{10}}{-4v^7}$ $-7v^3$

3. $\frac{2b^9c^{11}}{7bc^7}$ $\frac{2b^8c^4}{7}$

4. $\left(\frac{3s}{-4}\right)^2$ $\frac{9s^2}{16}$

5. $\frac{(2x^4y)^5}{-16xy^6}$ $-\frac{2x^{19}}{y}$

6. $\frac{(-3x^5y^3)^4 \cdot x^9y}{-9x^7y^2}$ $-9x^{22}y^{11}$

Melinda simplified the expression $\frac{(4a^8b^2)^3}{-8a^{12}b^3}$. Her equivalent expression after simplifying using the laws of exponents was $-8a^{12}b^3$.

7. Explain whether or not you agree with her expression.

I agree with her expression.

$$\frac{4^3a^{24}b^6}{-8a^{12}b^3} = \frac{64a^{12}b^3}{-8} = -8a^{12}b^3$$

8. Find and fix her mistake.

There is no mistake.

Students in Ms. Vale's 8th grade math class, had to simplify the algebraic expressions for homework. Each student made a mistake. Circle the students' errors and show your work to justify your answer.

9. Stefan's Work	10. Monique's Work
$\frac{(4r^4y)^2 \cdot r^5y^3}{2r^9y^2}$ $\frac{16r^6y^2 \cdot r^5y^3}{2r^9y^2}$ $\frac{8r^{11}y^5}{2r^9y^2}$ $4r^2y^3$	$\frac{(4r^4y)^2 \cdot r^5y^3}{2r^9y^2}$ $\frac{16r^8y^2 \cdot r^5y^3}{2r^9y^2}$ $\frac{8r^{40}y^6}{2r^9y^2}$ $4r^{31}y^4$
Correct Work: $\frac{16r^8y^2 \cdot r^5y^3}{2r^9y^2}$ $\frac{16r^{13}y^5}{2r^9y^2}$ $8r^4y^3$	Correct Work: $\frac{16r^8y^2 \cdot r^5y^3}{2r^9y^2}$ $\frac{16r^{13}y^5}{2r^9y^2}$ $8r^4y^3$